DSA 210 — Introduction to Data Science
Spring 2025–2026 | Project Proposal

Multi-Factor Predictors of Short-Term Rental Pricing: A Spatial Analysis of Airbnb Listings in New York City

Motivation

New York City hosts one of the most dynamic and complex short-term rental markets in the world. While basic listing characteristics (room type, number of reviews, availability) are commonly studied, the role of **urban convenience** and **neighborhood safety** as pricing drivers remains underexplored in combination. As a data science student, I aim to investigate whether access to public transit and local crime density jointly and independently explain price variation in Airbnb listings — beyond what property attributes alone can account for.

This project is particularly interesting because it requires fusing three distinct, heterogeneous datasets through spatial joins and feature engineering — a real-world data science challenge. The goal is to build a model that can identify price anomalies and quantify the premium (or discount) associated with subway proximity and neighborhood safety.

Data Sources

Three publicly available datasets will be used, each adding a distinct analytical layer:

1. NYC Airbnb Open Data

Source: Kaggle — New York City Airbnb Open Data

https://www.kaggle.com/datasets/dgomonov/new-york-city-airbnb-open-data

~48,895 listings with 16 features including price, room type, borough, latitude/longitude, number of reviews, and availability. This forms the primary dataset.

2. MTA Subway Stations

Source: Data.gov — MTA Subway Stations

https://catalog.data.gov/dataset/mta-subway-stations

GTFS-formatted geographic coordinates of all 496 NYC subway stations. Will be used to compute the *minimum distance from each Airbnb listing to the nearest station*, creating the key transit-access feature.

3. NYPD Complaint Data (Current Year-To-Date)

Source: NYC Open Data — NYPD Complaint Data

https://data.cityofnewyork.us/Public-Safety/NYPD-Complaint-Data-Current-Year-To-Date-/5uac-w243/data_preview

Real-time NYC crime complaint records, including complaint coordinates, borough, and offense severity (felony, misdemeanor, violation). Given the large volume (100MB+), only relevant columns (BORO_NM, LAW_CAT_CD, Latitude, Longitude) will be extracted and aggregated into neighborhood-level crime density scores.

Data Collection & Characteristics

All three datasets will be downloaded as CSV files and processed in Python using Pandas. The Airbnb dataset has approximately **48,895 samples** with **16 initial columns**. The NYPD dataset exceeds 100MB — feature selection will be applied immediately upon loading to retain only the four relevant columns, reducing memory footprint. The MTA station data is compact (~496 rows) and will be loaded in full.

A key enrichment step involves computing **minimum distance to subway** for each listing using the Haversine formula (via the *geopy* library), generating the engineered feature *min_dist_to_subway*. A second enrichment will assign each listing a **local crime density score** by spatial aggregation of NYPD complaints within a defined radius.

Planned Methodology

1. Preprocessing

Clean missing price values, handle outliers (e.g., listings priced at $0 or $10,000+), normalize borough/neighborhood names across datasets, and parse coordinate fields.

2. Enrichment (Spatial Join)

Link each Airbnb listing to its nearest subway station (Haversine distance) and compute a neighborhood-level crime density index from NYPD complaint counts.

3. Exploratory Data Analysis (EDA)

Visualize price distributions by borough, generate spatial heatmaps of price vs. subway proximity, and overlay crime hotspots using Matplotlib, Seaborn, and Folium.

4. Hypothesis Testing

Test whether listings within a 500-meter radius of a subway station command a statistically significant price premium, controlling for room type and borough (t-test / Mann-Whitney U).

5. Machine Learning Modeling

Train regression models (Linear Regression, Random Forest) on the enriched feature set to predict rental price. Evaluate using RMSE and R^2 with train/test splitting.